

A
PROJECT REPORT
ON
“Voice and Bluetooth Controlled Car”

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The project work, entitled

As **Voice and Bluetooth controlled Car** has been carried out at the establishment of

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DECLARATION

I hereby declare that the project work entitled, **Voice and Bluetooth controlled Car** submitted under guidance of Prof. **Prashant Malavadkar** , is my original work completed under the four walls of **Pimpri Chinchwad University**.

The Project Report submitted is my own work and has not been duplicated from any other source. I shall be responsible for any unpleasure moment/situation.

Aditya Santosh Wajge

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Aditya Santosh Wajge

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CHAPTER 1

INTRODUCTION

1.1 Introduction

The rapid advancement of technology in recent years has led to the development of intelligent systems that simplify human tasks and improve efficiency. Among these advancements, automation and wireless communication have become key areas of innovation. Robotics, combined with embedded systems and communication technologies, has opened new possibilities in various fields such as industrial automation, healthcare, defense, and home assistance. One such application is the development of a voice and Bluetooth controlled robotic car, which enables users to control a vehicle wirelessly using simple voice commands or mobile-based inputs. This project represents a practical implementation of modern technological concepts and demonstrates how everyday devices like smartphones can be integrated with embedded systems to perform real-time control operations.

The Voice and Bluetooth Controlled Car is designed to provide a user-friendly and efficient method of controlling a robotic vehicle without the need for physical interaction. The system uses an Arduino micro controller as the central processing unit, which is responsible for receiving and processing commands. Communication between the user and the vehicle is established through a Bluetooth module, typically the HC-05, which enables wireless data transmission between a smartphone and the robotic system. The user interacts with the system through an Android application that supports both manual button control and voice command input. Voice commands given by the user are converted into text using speech recognition software within the mobile device, and these commands are then transmitted to the Arduino via Bluetooth. Based on the received instructions, the Arduino controls the motors of the car, allowing it to move forward, backward, left, right, or stop.

This project highlights the importance of integrating hardware and software components to achieve a seamless control mechanism. The use of embedded systems such as Arduino makes the design cost-effective, flexible, and easy to implement. Additionally, the inclusion of wireless communication eliminates

the need for complex wiring and enhances the mobility of the system. The project also introduces the concept of human-machine interaction, where voice commands act as a natural and intuitive interface for controlling devices. This makes the system particularly useful for individuals who may have difficulty using traditional control mechanisms, such as elderly or physically challenged persons.

Furthermore, the system can be enhanced by integrating additional sensors and modules to increase its functionality. For example, ultrasonic sensors can be used for obstacle detection, allowing the vehicle to avoid collisions automatically. Similarly, cameras and other IoT-enabled devices can be incorporated for surveillance and monitoring applications. The project serves as a foundation for understanding advanced concepts such as automation, robotics, and Internet of Things (IoT), and it can be further developed into more sophisticated systems for real-world applications.

1.2 Abstract

In the age of automation and smart technologies, wireless control systems are gaining immense popularity for their efficiency and ease of use. This mini project titled "Voice and Bluetooth Controlled Car" aims to design and develop a prototype vehicle that can be operated using voice commands and Bluetooth connectivity. The project merges the concepts of embedded systems, wireless communication, and mobile application control to create an innovative and user-friendly control system.

The proposed system utilizes an Arduino microcontroller as the core processing unit. Bluetooth communication is established via the HC-05 module, allowing a smartphone to connect with the vehicle. The user can either give voice commands through a dedicated Android application or manually control the car using Bluetooth-enabled buttons on the app interface. Voice recognition is handled on the smartphone itself, converting spoken commands into serial data which is transmitted to the micro controller.

Based on the received input, the Arduino processes the instructions and drives the motors connected to the wheels, enabling the car to move forward, backward, left, right, or stop. The system eliminates the need for physical remotes, making operation more intuitive and hands-free. This project not only demonstrates the implementation of wireless control but also introduces basic elements of human machine interaction.

The integration of voice control makes the system more accessible

and paves the way for advanced applications such as smart robotics, assistive mobility solutions, and home automation. This project reflects a practical approach to understanding embedded systems, programming, and real-time control, offering valuable insight into the interdisciplinary nature of modern engineering problems

1.3 Existing System and Proposed System

In the field of robotics and automation, various systems have been developed over time to control robotic vehicles. These systems differ in terms of communication methods, complexity, cost, and usability. Understanding the limitations of existing systems helps in designing a more efficient and user-friendly solution. The proposed Voice and Bluetooth Controlled Car aims to overcome the drawbacks of traditional systems by introducing a simpler, more flexible, and cost-effective approach.

➤ Existing System

The existing systems for controlling robotic vehicles primarily include wired control systems, remote-controlled vehicles, and Wi-Fi-based control systems. Each of these systems has its own working mechanism and limitations.

Wired robot systems are one of the earliest methods used for controlling robots. In this system, the robot is connected to a controller using physical wires. The user sends commands through switches or control panels, which are transmitted via wires to the robot. Although this method is simple and reliable, it severely limits the movement of the robot due to the physical constraints of the wires. The robot cannot move freely beyond the length of the cable, making it unsuitable for large-scale or dynamic environments. Additionally, wires can get tangled or damaged, leading to operational issues.

Remote-controlled vehicles use RF (Radio Frequency) or infrared-based remote controllers to operate the robot. These systems eliminate the need for wires and provide better mobility compared to wired systems. The user can control the robot using a handheld remote device with buttons for different directions. However, these systems require a dedicated remote controller, which increases the overall cost. Moreover, the range of communication is often limited, and interference from other electronic devices can affect performance. Infrared-based systems also require a direct line of sight, which further restricts usability.

➤ **Proposed System**

The proposed system introduces a Voice and Bluetooth Controlled Car that overcomes the limitations of existing systems by combining simplicity, flexibility, and affordability. This system uses Bluetooth communication along with voice recognition technology to control the movement of the robot.

In the proposed system, a smartphone acts as the control device. The user gives voice commands through a mobile application, which uses built-in speech recognition to convert spoken words into text. These commands are then transmitted to the robot using a Bluetooth module (HC-05). The Arduino micro controller receives the commands and processes them to control the motors accordingly. Based on the command, the robot can move forward, backward, left, right, or stop.

Bluetooth-based communication offers several advantages over traditional methods. It eliminates the need for wires and provides sufficient range for short-distance applications. The setup is simple and does not require complex network configurations like Wi-Fi systems. Bluetooth modules are also cost-effective and easy to interface with micro-controllers, making them ideal for student projects and low-budget applications.

The integration of voice control adds another level of convenience to the system. Instead of using buttons or joysticks, the user can control the robot simply by speaking commands. This makes the system more intuitive and accessible, especially for elderly or physically challenged individuals. The absence of a physical remote reduces hardware dependency and makes the system more portable.

The proposed system also focuses on low-cost implementation. By using readily available components such as Arduino, Bluetooth module, and DC motors, the overall cost of the system is minimized without compromising functionality. The design is simple, easy to assemble, and requires minimal maintenance.

1.4 Scope and Objectives

➤ Scope

The scope of this project extends beyond the basic implementation of a voice-controlled robotic car. Although the current model is designed for short-range communication using Bluetooth, it can be further enhanced and adapted for a wide range of real-world applications.

One of the major areas where this system can be applied is in military surveillance. In defense applications, robots are often used to access areas that are dangerous or inaccessible to humans. The voice-controlled robotic car can be equipped with cameras and sensors to monitor enemy zones, detect obstacles, or carry out reconnaissance missions. Since the system can be controlled wirelessly, it reduces the risk to human life and improves operational efficiency in critical situations.

Another important application lies in home automation. The system can be modified to control household devices or perform daily tasks using voice commands. For example, it can be integrated with smart home systems to operate appliances, transport objects within a house, or assist elderly individuals in their daily activities. The use of voice control makes the system highly user-friendly and suitable for people with limited mobility.

➤ Objectives

The primary objective of this project is to design and develop a wireless robotic car that can be controlled efficiently using modern communication technologies. The system is intended to eliminate the need for physical control mechanisms and provide a more flexible and user-friendly method of operation.

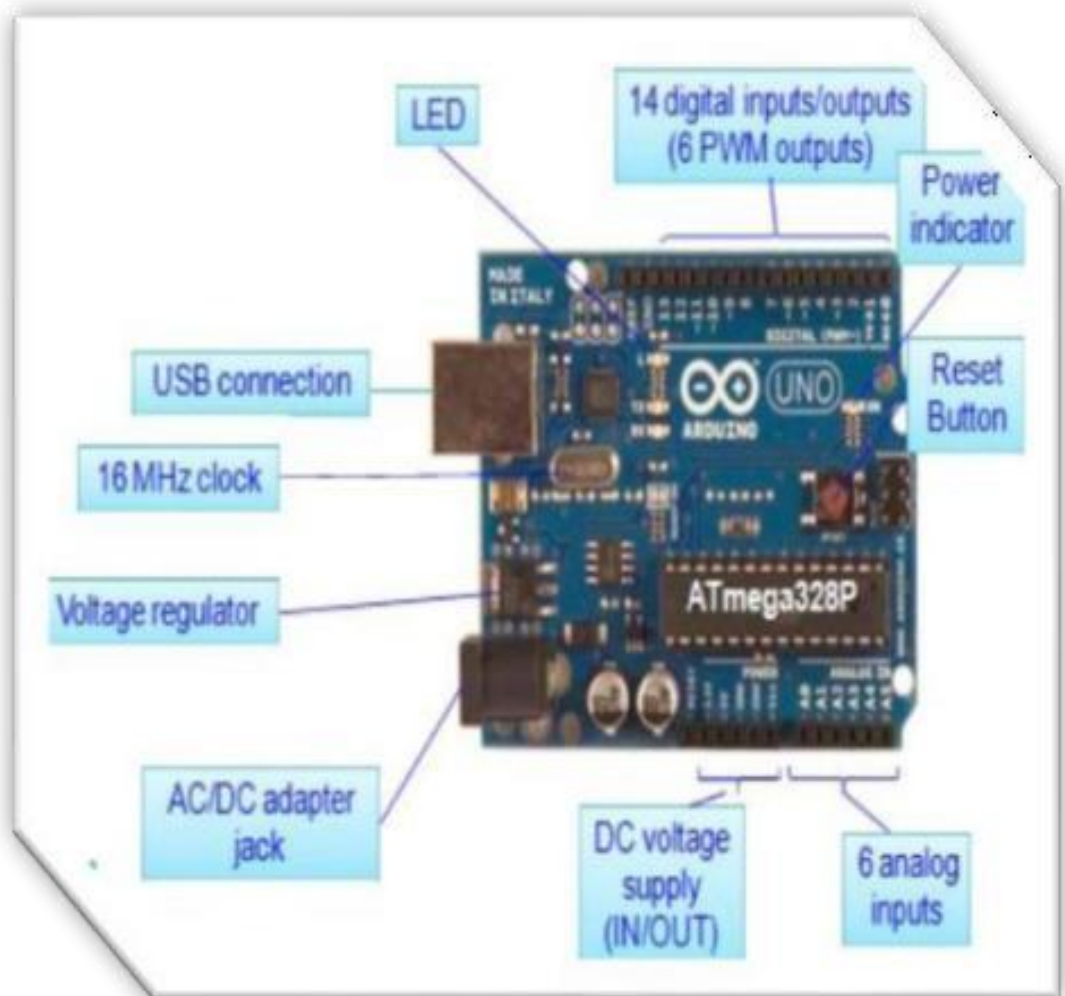
One of the key objectives is to design a wireless robotic car that operates without any physical connection between the user and the system. Traditional wired systems restrict the movement of robots and limit their usability. By implementing wireless communication, the robot can move freely within the range of the communication module, making it suitable for real-time applications. The use of Bluetooth technology ensures stable and reliable communication over short distances while maintaining simplicity in design.

Another important objective is to implement voice recognition control in the system. Voice control provides a natural and intuitive way of interacting with machines. Instead of relying on manual input devices such as buttons or joysticks, the user can simply give voice commands to control the movement of the robot. This not only enhances user convenience but also improves accessibility, especially for elderly and physically challenged individuals. The system utilizes speech recognition capabilities available in smartphones to convert voice commands into text, which is then transmitted to the robot for execution.

1.5 Operating Environment - Hardware and Software

➤ Arduino UNO ~

Arduino Uno is a microcontroller board based on the ATmega328P. It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header and a reset button.



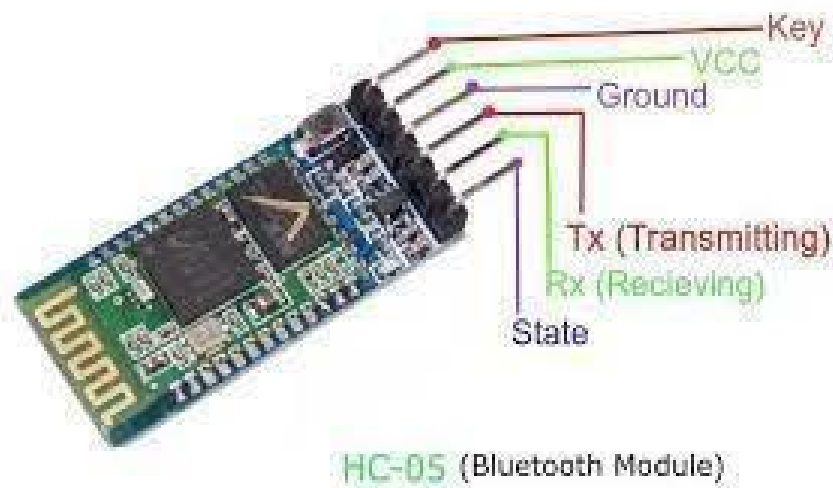
➤ **L293D Motor Driver Shield~**

The brains of the shield are two L293D motor drivers and a 74HC595 shift register. The L293D is a dual-channel H-Bridge motor driver that can control two DC motors or a single stepper motor. Because the shield includes two such motor drivers, it can control up to four DC motors or two stepper motor.



➤ HC-05 Bluetooth module ~

The HC-05 Bluetooth module is a wireless communication device used to establish a connection between the Arduino and a smartphone or other Bluetooth-enabled devices. It operates on the Bluetooth 2.0 protocol and supports serial communication (UART), making it easy to interface with microcontrollers. The module can work in both master and slave modes, though it is commonly used in slave mode for receiving commands from a mobile device.



➤ **HC – SR04 Ultrasonic Sensor ~**

The HC-SR04 ultrasonic sensor is used for measuring distance and detecting obstacles in the robotic system. It works on the principle of ultrasonic wave transmission and reflection. The sensor emits ultrasonic waves through the trigger pin, and when these waves hit an object, they are reflected back to the sensor and received through the echo pin. The time taken for the signal to return is used to calculate the distance between the sensor and the object. It operates on a 5V power supply and provides accurate distance measurements within a range of 2 cm to 400 cm.



➤ Servo Motor ~

A servo motor is a type of actuator used for precise control of angular position. Unlike regular DC motors, a servo motor can rotate to a specific angle, usually between 0° and 180° , based on the control signal it receives. It operates using Pulse Width Modulation (PWM), where the width of the pulse determines the angle of rotation.



➤ **Motor ~**

A DC motor is any of a class of rotary electrical motors that converts direct current electrical energy into mechanical energy. The most common types rely on the forces produced by magnetic fields.



➤ **Power Supply ~**

The car is powered by two high quality and durable lithium-ion batteries. These batteries each, have a capacity of 6800mAh and a voltage of 4.2V.



1.6 Technology Used:-

IoT Introduction

Internet of Things (IoT) connects devices to enable communication and automation.

In this project, IoT is implemented using a simple communication flow where devices interact wirelessly.

Architecture:-

- Smartphone → Bluetooth → Arduino → Motors

Explanation:

- The smartphone sends voice commands
- Bluetooth module (HC-05) transmits the data
- Arduino processes the command
- Motors execute the movement

Operating Systems:-

Windows:

- Used for development and programming
- Arduino IDE runs on Windows

Android:

- Used for controlling the robot
- Sends voice commands via Bluetooth

CHAPTER 2

PROPOSED SYSTEM

2.1 Previous Work / Research

Akshay Bhati et al. [1] Authors developed a smartphone-controlled robot car. In this project, Arduino UNO is used as a central component, to which all the other components are interfaced. The designed vehicle is controlled wirelessly through a smartphone with the help of a Wi-Fi module. On detecting an obstacle, a notification is sent to the smartphone and the robot performs pick and place operation with the help of a robotic arm mounted onto the vehicle.

H. Rissanen et. al [2] The author has developed a planned vehicle that is controllable by utilizing a Bluetooth module. A mechanical knock sensor is connected on the facade of the vehicle to decide whether an impact has happened to it, and likewise, gives the specific time of crash.

S. Mandal et. al. [3] Authors designed a Robot Vehicle to follow a black line path while accommodating various other features such as collision detection and avoidance or falling from a certain height with great stability and control. The design is incorporated with IR sensors, Bluetooth and Wi-Fi modules interfaced with a central Microcontroller Arduino UNO and is controlled by user through long range Wi-Fi connectivity. The path of the car can be changed by the user.

Ritika Pahuja et. al. [4] They have elaborated how to control robot-controlled car using Wi-Fi module through android application of an android mobile phone. The robot has a smart phone attached to it. It captures the video in front of robot. Robot contains Bluetooth Compiler, Controllers, driver, IR, ADC.

Mrumal.K. Pathak et.al [5] The motivation behind this paper is to furnish amazing computational android stages with less difficult robot equipment design. This paper depicts how to control a robot utilizing portable through Bluetooth communication, a few highlights about Bluetooth innovation, segments of the versatile and robot.

S R Madkar (Assistant Professor), Vipul Mehta, et.al [6] Authors provide powerful computational android platforms with simpler robot hardware architecture. It describes how to control a robot using mobile through Bluetooth communication, some features about Bluetooth technology, components of the mobile and robot.

Aniket R. Yeole, et.al [7] The authors have designed a robot that can be controlled using an application running on an android phone. It sends control command via Bluetooth which has certain features like controlling the speed of the motor, sensing and sharing the information with phone about the direction and distance of the robot from the nearest obstacle.

T. L. Chien, H. Guo, et.al [8] Authors describes a robot that extinguishes fire in real time is designed to extinguish the fire in the event of a fire disaster. It consists of two fire sensors that sense fire and then relay the signal to the module for motion control. The robot is then designated using a modern GSM to communicate using an RS232 interface and move to the fire location and extinguish the fire and also alarm the customer via a cell phone. To ease the movement of the robot, sets of IR sensors are used to facilitate the movement in free space and to avoid obstacles.

Prof. Bhuvaneshwari Jolad et.al [9] The authors define how to recognize voice of Robot:

- Small vocabulary systems
- Medium vocabulary systems
- Large vocabulary systems. They also define different types of speaker systems
- Speaker dependent systems
- Speaker independent systems
- Speaker adaptive systems .

R. M. Narayana, et.al [11] Authors develop the remote buttons in the android

app which control the robot motion with them. Adding to it, Bluetooth

communication is used to interface controller and android. Controller is interfaced to the Bluetooth module through UART protocol.

Jorge Kazacos Winter [12] The main aim of this project is the transfer of information wirelessly between a smartphone and the robot and developing the robot and its communication system underneath a low price and open-source philosophy. 3D design technique is used to style the structure of the robot with the facilitation of parametrical modelling software.

M. Selvam [13] has projected design to develop a robotic system which has a wireless camera and night vision camera attached to it for surveillance. Bluetooth is implemented in the project for providing connection between robot and smartphone.

Vito M. Guardì [14] The author has evolved the method of Bluetooth technology by developing an android app for a robot which is driven by a microcontroller. The central idea of his work is to show that one android app can be operated using totally different electronic devices.

Ranjith Kumar Goud et.al [15] Authors invented a pick and drop robot for diffusing a bomb remotely with safety. For the robotic arm, they used a pair of motors and another pair as the wheels of the robot for controlling the movement. Connectivity is established using Bluetooth. The micro-controller used is LPC2148. They had also attached a wireless camera for remote surveillance.

Arpit Sharma et.al [16] The authors designed the robot vehicle in which phone uses motion sensors and records the gestures sent via an android mobile phone. It also has an inbuilt accelerometer and Bluetooth module for controlling the movements of a robot.

Zhao Wang, Eng Gee Lim, et.al [17] designed a mechanical vehicle that is structured with the assistance of a Microcontroller Arduino UNO. The vehicle is

wirelessly controlled with the help of a Bluetooth module. On detecting any object within a threshold distance, the car stops and analyses all directions. Whichever provides more clearance, the car moves in that direction.

N. Firthous Begum, et.al [18] Authors talks about a pick and place robotic car which is designed with a purpose of picking the obstacle in front of it and placing aside. The car is controlled via Java programmed application and is equipped with a camera, placed on the car for surveillance.

Xiao Lu, et.al [19] The authors developed a model in which the smartphone is capable of IFLYTEK voice as well as handwritten input. The design is therefore robust, suitable, and practical for use and it also ensures the reliability of the full system. Wi-Fi serves as the mode for connectivity between the smartphone and robot and it makes it easy and absolutely convenient for controlling the robot so that it can act according to the commands.

Zhenjun He, et.al [20] Authors describes a mine detection robot with the help of different sensors. For detection purposes, a combination of ultrasonic sensor, gas sensor, temperature sensor, humidity sensor is used. The system is designed to assist the military in places where mine detection is highly probable. Further, the designed car is accompanied with a camera for video surveillance and an alert is sent to a smartphone with the help of IOT whenever a mine is detected.

➤ **Selected paper from Literature Survey**

Akshay Bhati et al. [1] Authors developed a multi-purpose smartphone-controlled robot car. In this project, Arduino UNO is used as a central component, to which all the other components are interfaced. The designed vehicle is controlled wirelessly through a smartphone with the help of a Wi-Fi module. On detecting an obstacle, a notification is sent to the smartphone and the robot performs pick and place operation with the help of a robotic arm mounted onto the vehicle.

➤ **Overcome the Literature**

The author uses a Wi-Fi module to control the movement of the car. We'll be using a Bluetooth module in our project so as to control the movement of the car with help of our voice commands.

➤ **Motivation**

The project has utilized a Wi-Fi module to interact and move the car as per the user's need which can be tedious.

➤ **Problem Definition**

In this project, we will learn how to make Voice Controlled Robot Car using Arduino. The robotic car can be controlled wirelessly via voice commands directly from the user. The robot can move forward, backward, left, and right and can also be stopped.

CHAPTER 3

PROPOSED IOT MODEL DESIGN AND IMPLEMENTATION

3.1 Choice of Sensors and Actuators in Proposed IoT Model

➤ Sensors Used

Ultrasonic Sensor (HC-SR04):

The ultrasonic sensor is used for obstacle detection and distance measurement. It works by transmitting ultrasonic waves and receiving the reflected signal from nearby objects. Based on the time taken for the echo to return, the distance is calculated. This sensor helps the robotic car avoid collisions and enhances safety during operation.

➤ Reason for Selection:

- Low cost and easily available
- High accuracy for short-distance measurement
- Simple interfacing with Arduino

➤ Actuators Used

DC Motors:

DC motors are used to drive the wheels of the robotic car. They convert electrical energy into mechanical motion, enabling movement in different directions such as forward, backward, left, and right.

Motor Driver (L293D):

The motor driver acts as an interface between the Arduino and the DC motors. It controls the direction and speed of the motors based on signals received from the Arduino. It also protects the microcontroller from high current.

➤ Reason for Selection:

- Efficient control of motor direction and speed
- Ability to drive multiple motors
- Reliable and widely used in robotics

3.2 Working of IoT Model

The proposed system operates using Bluetooth-based voice control.

➤ **Working Process:**

1. The user gives a voice command through an Android application.
2. The application converts the voice input into a text command.
3. The command is transmitted to the Arduino via the Bluetooth module (HC-05).
4. The Arduino receives the command through serial communication.
5. Based on the command, the Arduino processes and sends control signals to the motor driver (L293D).
6. The motor driver activates the DC motors to perform the required movement.
7. Simultaneously, the ultrasonic sensor continuously monitors for obstacles.
8. If an obstacle is detected within a certain range, the system can stop or change direction to prevent collision.

This ensures both user control and safety during operation.

3.3 Design of Proposed IoT Model

The system is designed as a simple and efficient IoT-based robotic model.

Design Components:

- **Input:** Voice commands from the Android application
- **Processing Unit:** Arduino UNO microcontroller
- **Communication Module:** HC-05 Bluetooth module
- **Output:** DC motors for movement
- **Safety Component:** Ultrasonic sensor for obstacle detection

Design Features:

- Wireless communication using Bluetooth
- Real-time response to user commands
- Modular structure allowing easy upgrades
- Cost-effective and easy to implement

3.4 Flow of Proposed IoT Model

The system follows a sequential flow for operation.

Flow Steps:

1. The system is powered on and initialized.
2. Arduino initializes all connected components (Bluetooth module, sensors, motor driver).
3. The system waits for input from the user.
4. The user sends a voice command via the Android application.
5. The Bluetooth module receives and forwards the command to Arduino.
6. Arduino processes the command.
7. The ultrasonic sensor checks for obstacles:
 - If an obstacle is detected, the system stops or avoids it.
 - If no obstacle is detected, the command is executed normally.
8. Arduino sends signals to the motor driver.
9. Motors perform the required movement.
10. The process repeats until the system is turned off.

CHAPTER 4 ARCHITECTURE OF SMART SYSTEM

4.1 Overview of Proposed Smart System

The proposed smart system is a **Bluetooth and voice-controlled robotic car** designed using IoT principles. It enables wireless communication between a user and a robotic device for efficient control and automation.

The system allows the user to send commands through an Android application using voice input. These commands are transmitted via Bluetooth to the Arduino microcontroller, which processes the instructions and controls the movement of the robot accordingly.

The smart system integrates multiple components such as a microcontroller, communication module, sensors, and actuators to perform real-time operations. The ultrasonic sensor enhances the system by providing obstacle detection, ensuring safe navigation.

Key Features of the Smart System:

- Wireless control using Bluetooth
- Voice-based command system
- Real-time processing and response
- Obstacle detection for safety
- Low-cost and easy-to-use design

The system is designed to be flexible and can be extended for applications such as surveillance, automation, and assistive technologies.

4.2 Description of IoT Proposed Model Architecture

The architecture of the proposed IoT model is based on a layered approach that ensures smooth communication between different components of the system.

1. Input Layer (User Interface Layer):

This layer consists of the Android application through which the user interacts with the system. The user provides voice commands, which are converted into text instructions and sent to the communication layer.

2. Communication Layer (Network Layer):

This layer is responsible for transmitting data between the user device and the hardware system. In this project, the HC-05 Bluetooth module is used to establish a wireless connection between the smartphone and the Arduino. It ensures reliable and real-time data transfer.

3. Processing Layer (Control Layer):

The Arduino UNO acts as the central processing unit. It receives commands from the Bluetooth module, interprets them, and makes decisions based on the input and sensor data. It also controls the overall operation of the system.

4. Execution Layer (Actuator Layer):

This layer includes the motor driver (L293D) and DC motors. Based on the processed commands, the motor driver controls the motors to perform actions such as moving forward, backward, or turning.

5. Sensing Layer (Feedback Layer):

The ultrasonic sensor continuously monitors the surroundings of the robot. It provides feedback to the Arduino about obstacles, allowing the system to stop or adjust movement to avoid collisions.

Working of Architecture:

1. The user gives a voice command through the Android application.
2. The command is transmitted via Bluetooth.
3. Arduino receives and processes the command.
4. Sensor data is checked for safety conditions.
5. Appropriate signals are sent to actuators.
6. The robot performs the desired action.

CHAPTER 5

IOT MODEL DETAILS

5.1 Choice of IoT Platforms and Integrated Tools

In the proposed system, IoT platforms play a crucial role in controlling and managing hardware components.

➤ **Arduino Platform**

The system primarily uses the Arduino UNO as the main IoT platform. Arduino is a microcontroller-based development board used for real-time control applications.

➤ **Features:**

- Based on ATmega328P microcontroller
- Easy to program using Arduino IDE
- Supports digital and analog input/output pins
- Suitable for embedded and real-time systems

5.2 IoT Devices in Proposed System

The proposed system consists of several IoT devices that work together to perform the required operations.

List of Devices:

- Arduino UNO (processing unit)
- HC-05 Bluetooth module (communication device)
- Ultrasonic sensor (distance sensing device)
- Motor driver L293D (control device)
- DC motors (movement actuators)
- Android smartphone (user interface device)

Functionality:

- These devices are interconnected to enable communication, processing, sensing, and execution.

- The smartphone sends commands, Arduino processes them, and actuators perform actions.
- Sensors provide feedback to ensure safe and efficient operation.

5.3 Description of Input and Output Components

➤ Input Components

1. Ultrasonic Sensor:

- Used to detect obstacles and measure distance
- Sends data to Arduino for processing

2. Bluetooth Module (HC-05):

- Receives input commands from the Android device
- Transfers data to Arduino via serial communication

3. Smartphone (Voice Input):

- Acts as the primary input interface
- Converts voice commands into text instructions

➤ Output Components

1. DC Motors:

- Responsible for movement of the robotic car
- Execute commands like forward, backward, left, and right

2. Motor Driver (L293D):

- Controls speed and direction of motors
- Acts as an interface between Arduino and motors

➤ **Supporting Components**

1. USB Cable:

- Used to upload code from computer to Arduino
- Provides power during programming

2. Processor (Microcontroller - ATmega328P):

- Core processing unit of Arduino
- Executes program instructions and controls all operations

3. Micro Electronic Chips / IC Chips:

- Includes L293D motor driver IC
- Helps in signal processing and motor control

4. Boards:

- Arduino UNO board serves as the main circuit platform
- Provides connections for all components

5. Power Supply:

- Supplies electrical energy to the system
- Typically uses batteries (e.g., 3.7V lithium batteries)
- Ensures continuous operation of motors and sensors

CHAPTER 6

**DEVELOPMENT AND IMPLEMENTATION
OF MODEL**

6.1 Code Implementation:-

➤ Bluetooth Control Code:-

```
#include <AFMotor.h>
#include <SoftwareSerial.h>

SoftwareSerial bluetoothSerial(9, 10); // RX, TX

AF_DCMotor motor1(1, MOTOR12_1KHZ);
AF_DCMotor motor2(2, MOTOR12_1KHZ);
AF_DCMotor motor3(3, MOTOR34_1KHZ);
AF_DCMotor motor4(4, MOTOR34_1KHZ);

char command = 'S'; // default stop

void setup()
{
    bluetoothSerial.begin(9600);
}

void loop()
{
    if (bluetoothSerial.available() > 0)
    {
        command = bluetoothSerial.read();
    }

    switch (command)
    {
        case 'F':
            forward();
            break;
        case 'B':
            back();
```

```
        break;
    case 'L':
        left();
        break;
    case 'R':
        right();
        break;
    case 'S':
        Stop();
        break;
    }
}

void forward()
{
    motor1.setSpeed(255);
    motor2.setSpeed(255);
    motor3.setSpeed(255);
    motor4.setSpeed(255);

    motor1.run(FORWARD);
    motor2.run(FORWARD);
    motor3.run(FORWARD);
    motor4.run(FORWARD);
}

void back()
{
    motor1.setSpeed(255);
    motor2.setSpeed(255);
    motor3.setSpeed(255);
    motor4.setSpeed(255);
```

```
    motor1.run(BACKWARD);  
    motor2.run(BACKWARD);  
    motor3.run(BACKWARD);  
    motor4.run(BACKWARD);  
}
```

```
void left()  
{  
    motor1.setSpeed(200);  
    motor2.setSpeed(200);  
    motor3.setSpeed(200);  
    motor4.setSpeed(200);
```

```
    motor1.run(BACKWARD);  
    motor2.run(BACKWARD);  
    motor3.run(FORWARD);  
    motor4.run(FORWARD);  
}
```

```
void right()  
{  
    motor1.setSpeed(200);  
    motor2.setSpeed(200);  
    motor3.setSpeed(200);  
    motor4.setSpeed(200);
```

```
    motor1.run(FORWARD);  
    motor2.run(FORWARD);  
    motor3.run(BACKWARD);  
    motor4.run(BACKWARD);  
}
```

```
void Stop()
```

```

{
  motor1.run(RELEASE);
  motor2.run(RELEASE);
  motor3.run(RELEASE);
  motor4.run(RELEASE);
}

```

➤ **Voice Control Code:-**

```

#include <AFMotor.h>
#include <SoftwareSerial.h>

SoftwareSerial bluetoothSerial(9, 10); // RX, TX

AF_DCMotor motor1(1, MOTOR12_1KHZ);
AF_DCMotor motor2(2, MOTOR12_1KHZ);
AF_DCMotor motor3(3, MOTOR34_1KHZ);
AF_DCMotor motor4(4, MOTOR34_1KHZ);

String command = "";
char currentState = 'S'; // Default STOP

void setup()
{
  bluetoothSerial.begin(9600);
}

void loop()
{
  // Read voice command
  while (bluetoothSerial.available())
  {
    char c = bluetoothSerial.read();

```



```

    command += c;
    delay(5);
}

if (command.length() > 0)
{
    command.trim();
    if (command == "forward") currentState = 'F';
    else if (command == "back") currentState = 'B';
    else if (command == "left") currentState = 'L';
    else if (command == "right") currentState = 'R';
    else if (command == "stop") currentState = 'S';
    command = "";
}

// Execute continuously
switch (currentState)
{
    case 'F':
        forward();
        break;

    case 'B':
        back();
        break;

    case 'L':
        forwardLeft(); //    changed
        break;

    case 'R':
        forwardRight(); //    changed
        break;

```

```

        case 'S':
            Stop();
            break;
    }
}

// ===== FUNCTIONS =====

void forward()
{
    setSpeed(180, 180);
    motor1.run(FORWARD);
    motor2.run(FORWARD);
    motor3.run(FORWARD);
    motor4.run(FORWARD);
}

void back()
{
    setSpeed(180, 180);
    motor1.run(BACKWARD);
    motor2.run(BACKWARD);
    motor3.run(BACKWARD);
    motor4.run(BACKWARD);
}

// LEFT TURN (forward + slow left side)
void forwardLeft()
{
    setSpeed(100, 200); // left slow, right fast
    motor1.run(FORWARD);
    motor2.run(FORWARD);
    motor3.run(FORWARD);

```

```

    motor4.run(FORWARD);
}

//    RIGHT TURN (forward + slow right side)
void forwardRight()
{
    setSpeed(200, 100); // left fast, right slow
    motor1.run(FORWARD);
    motor2.run(FORWARD);
    motor3.run(FORWARD);
    motor4.run(FORWARD);
}

void Stop()
{
    motor1.run(RELEASE);
    motor2.run(RELEASE);
    motor3.run(RELEASE);
    motor4.run(RELEASE);
}

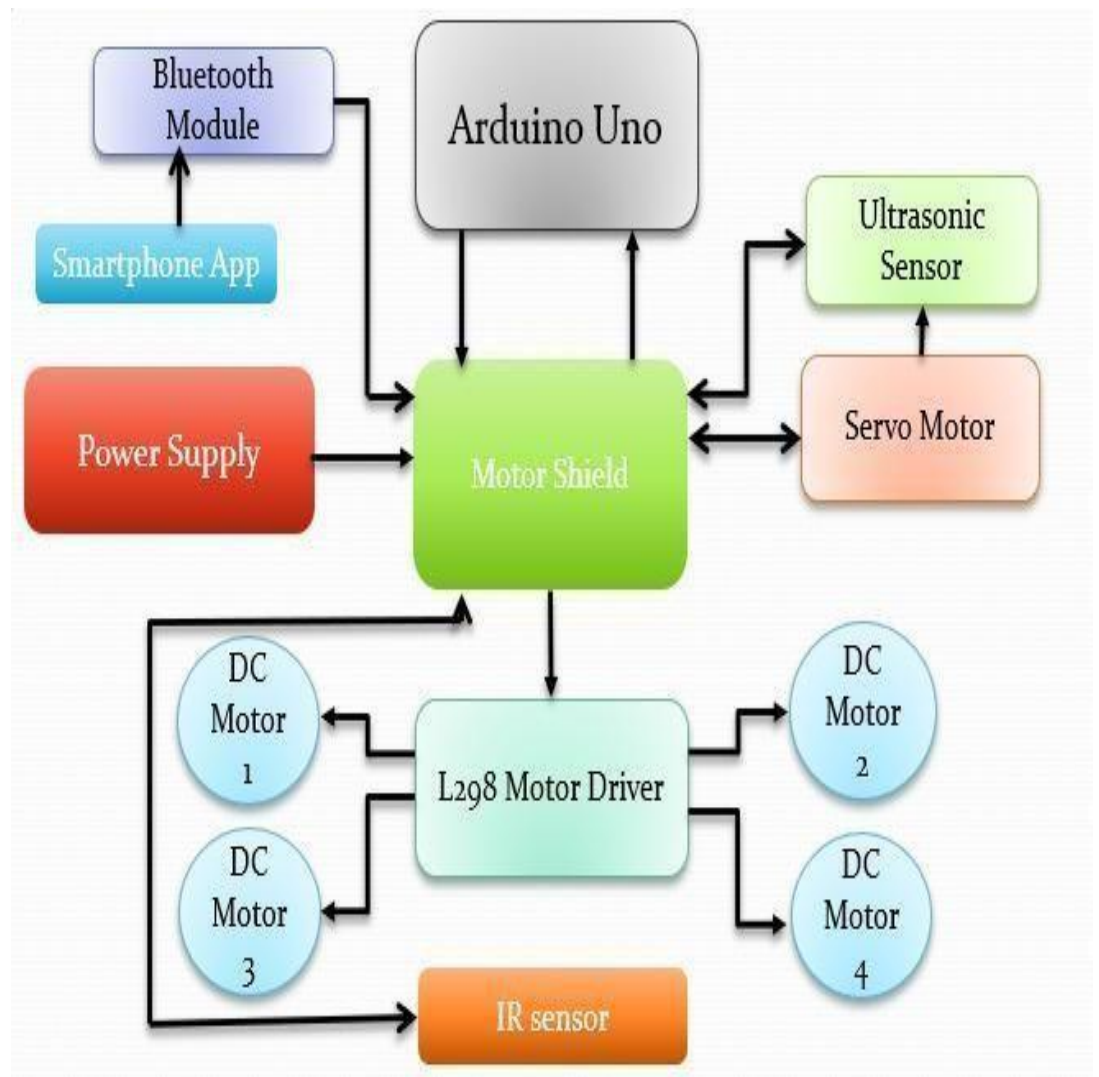
// Custom speed: left side motors & right side motors
void setSpeed(int leftSpeed, int rightSpeed)
{
    // Left motors
    motor1.setSpeed(leftSpeed);
    motor2.setSpeed(leftSpeed);

    // Right motors
    motor3.setSpeed(rightSpeed);
    motor4.setSpeed(rightSpeed);
}

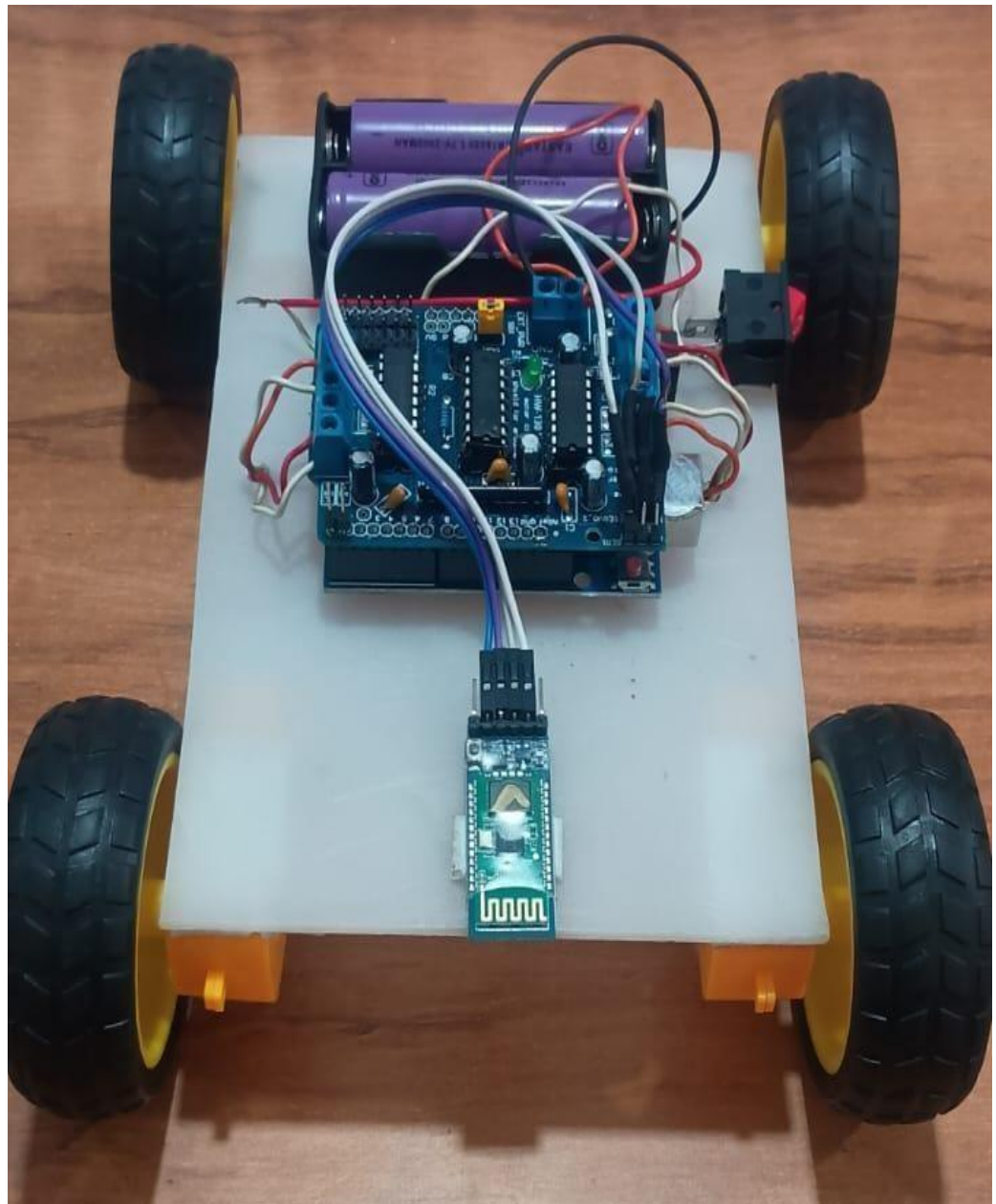
```

6.2 Screenshots and Diagrams:-

➤ Block Diagram:-

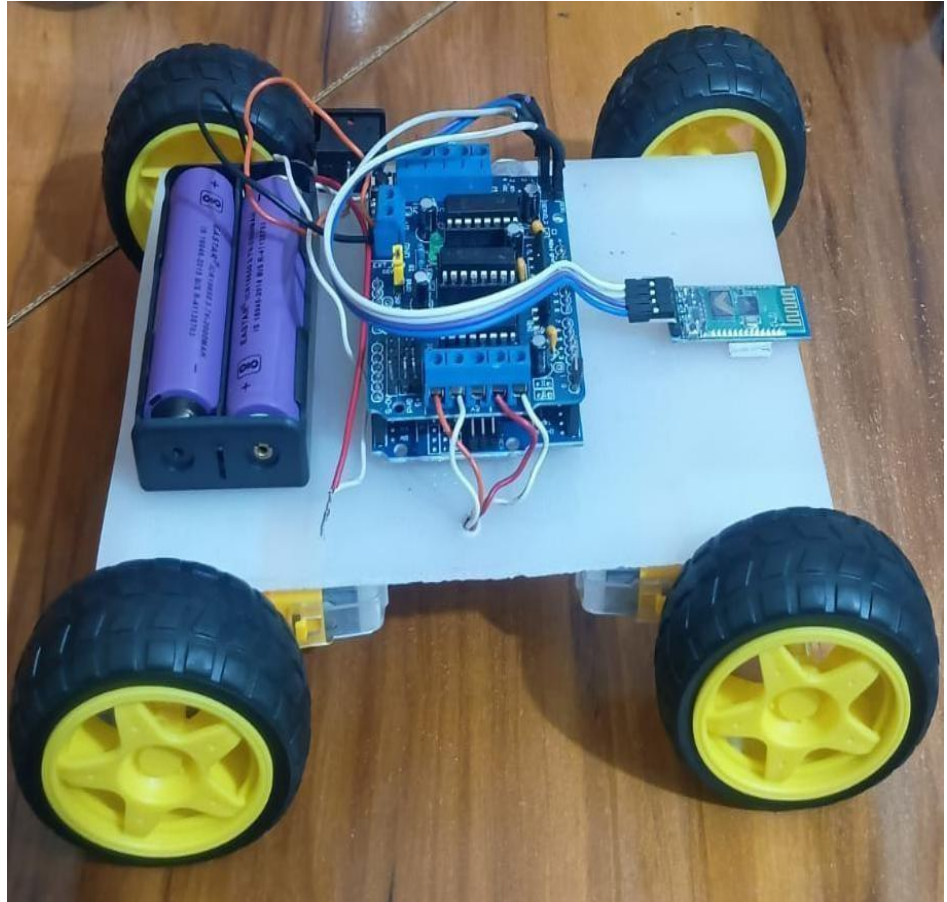


➤ **Actual Model:-**

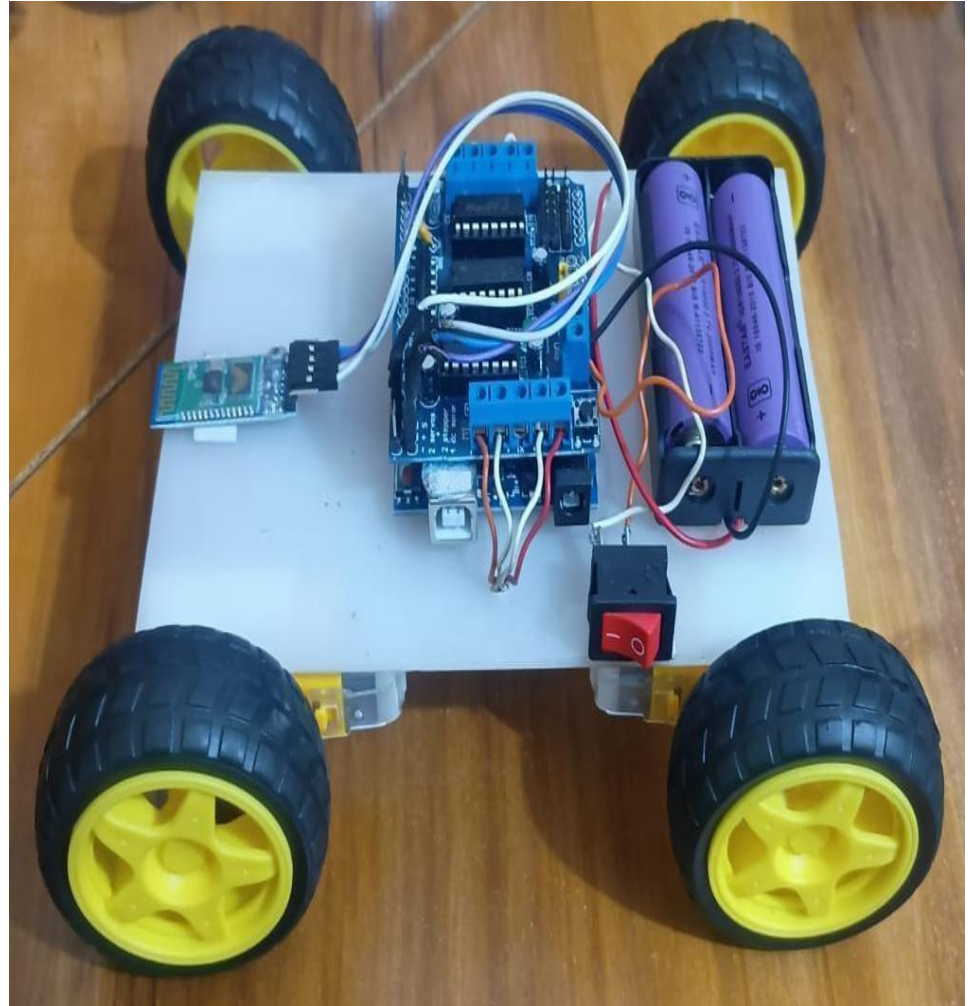




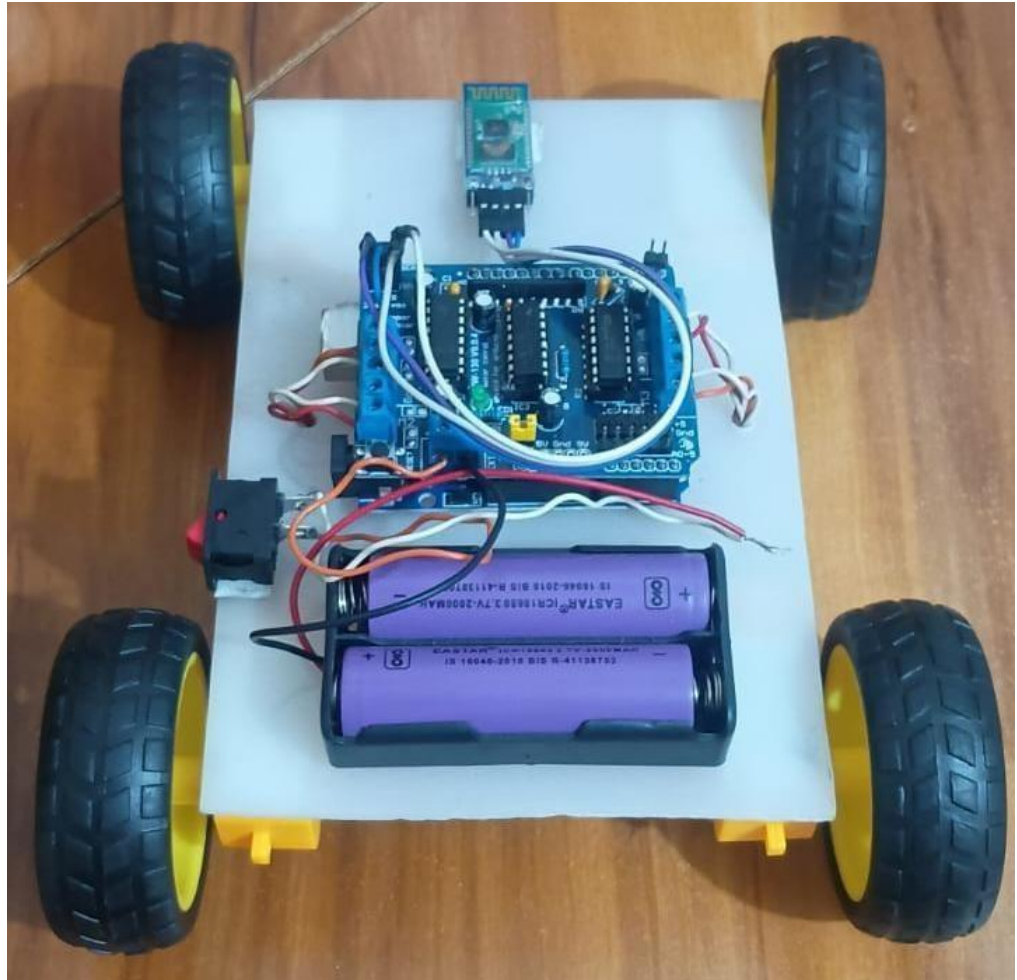
Left side



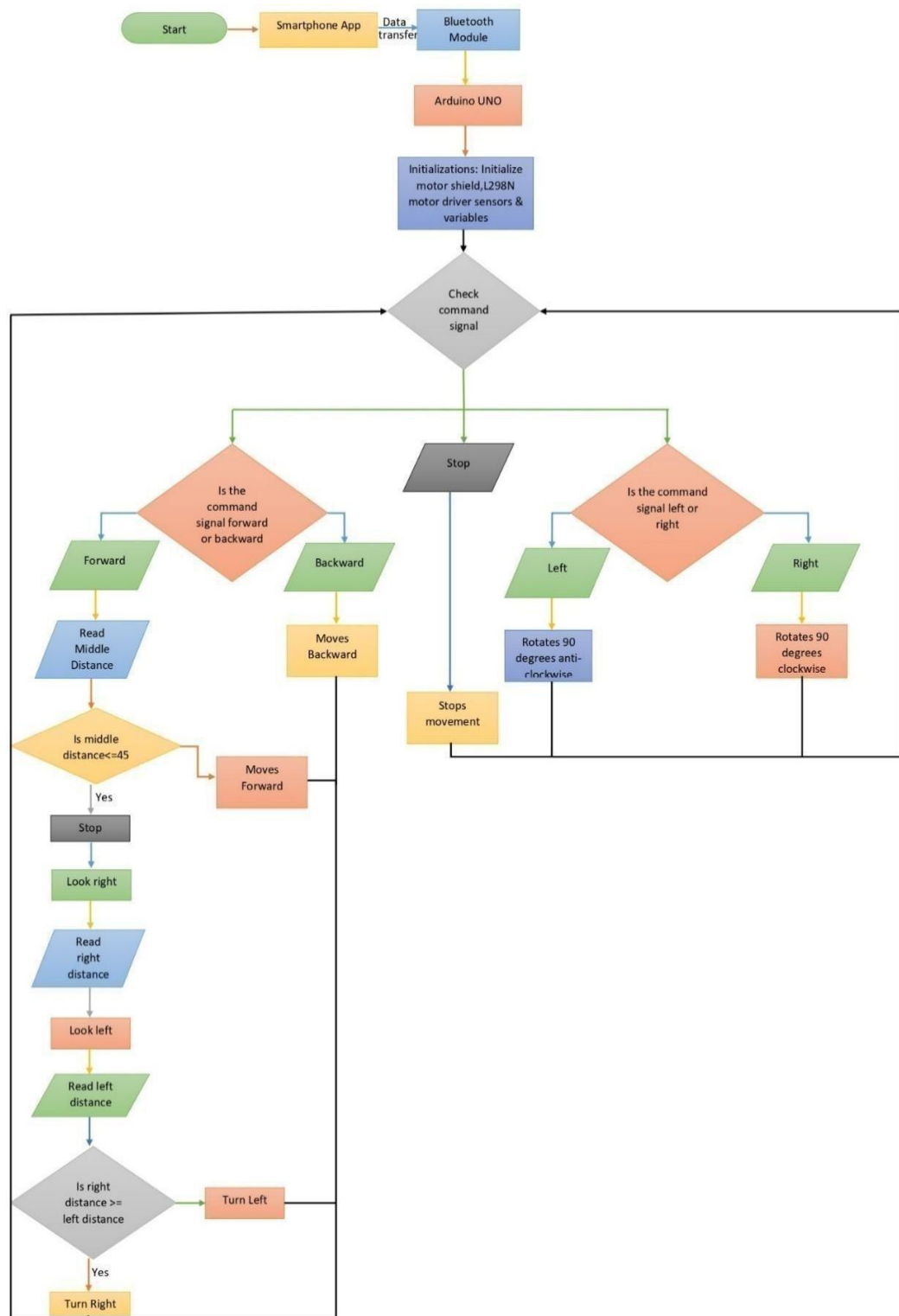
➤ **Right Side**



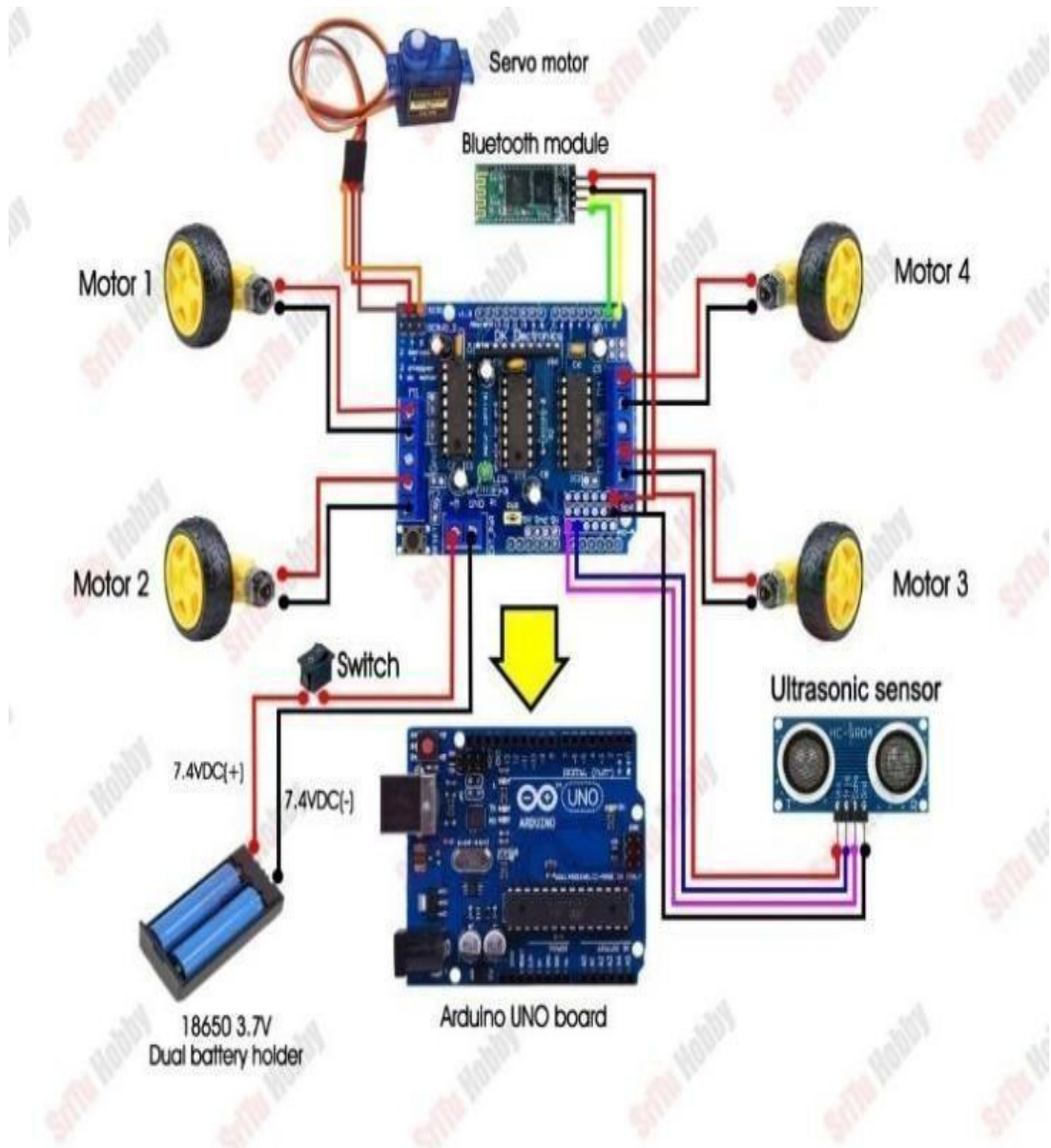
➤ **Back Side**



➤ Flow Chart



➤ **Circuit Diagram**



CHAPTER 7

TESTING OF IOT MODEL

7.1 Testing Strategy and Methodology

The testing strategy for the proposed IoT model focuses on verifying the functionality, reliability, and performance of each component as well as the complete system.

➤ Testing Strategy:

- **Unit Testing:**

Each component such as Arduino, Bluetooth module, ultrasonic sensor, and motors is tested individually to ensure proper functioning.

- **Integration Testing:**

After individual testing, components are combined to verify proper communication and coordination between modules.

- **System Testing:**

The entire system is tested as a whole to check overall performance and real-time response.

- **User Testing:**

The system is tested with actual user inputs (voice commands) to evaluate usability and responsiveness.

➤ Methodology:

1. Identify all hardware and software components.
2. Test each component independently.
3. Integrate components step by step.
4. Execute test cases for different commands (forward, backward, etc.).
5. Observe system behavior and record results.
6. Identify and fix errors.
7. Repeat testing until desired performance is achieved.

This systematic approach ensures that the system operates efficiently and reliably.

7.2 Testing of Sensors and Actuators

➤ Testing of Sensors

Ultrasonic Sensor:

- Tested by placing objects at different distances.
- Verified accuracy of distance measurement.
- Checked whether the system detects obstacles correctly.
- Ensured that the robot stops or reacts when an obstacle is detected within a set range.

IR Sensor (if used):

- Tested for object detection capability.
- Verified response to nearby obstacles or surfaces.

➤ Testing of Actuators

DC Motors:

- Tested for correct movement (forward, backward, left, right).
- Verified speed and smooth operation.
- Checked response to different commands.

Motor Driver (L293D/L298):

- Tested for proper control of motor direction.
- Verified that it correctly receives signals from Arduino.
- Ensured no overheating or malfunction during operation.

Servo Motor (if used):

- Tested for correct angle rotation.
- Verified response to control signals from Arduino.

7.3 Testing of Proposed IoT Model

The complete IoT system is tested to evaluate overall performance and integration.

➤ **Test Scenarios:**

- Sending voice commands through the Android app
- Verifying Bluetooth connectivity and data transmission
- Checking real-time response of the robot
- Observing movement accuracy based on commands
- Testing obstacle detection and automatic stopping

➤ **Performance Evaluation:**

- **Accuracy:** Commands are executed correctly
- **Response Time:** Minimal delay between input and action
- **Reliability:** Stable Bluetooth connection and consistent operation
- **Safety:** Proper obstacle detection and avoidance

CHAPTER 8

CONCLUSION

8.1 Summary

The proposed system is a Bluetooth and voice-controlled robotic car based on IoT concepts. It integrates hardware components such as Arduino UNO, HC-05 Bluetooth module, ultrasonic sensor, motor driver, and DC motors to achieve wireless control and automation.

The system allows users to control the robot using voice commands through an Android application. The commands are transmitted via Bluetooth, processed by the Arduino, and executed through motors. The ultrasonic sensor enhances the system by detecting obstacles and ensuring safe operation.

The project successfully demonstrates:

- Wireless communication between devices
- Real-time control and response
- Integration of sensors and actuators
- Implementation of IoT-based automation

Overall, the system is efficient, cost-effective, and user-friendly, making it suitable for various real-world applications.

8.2 Limitations and Challenges

Despite its effectiveness, the system has certain limitations and challenges:

- Limited Range:
Bluetooth communication typically works within a short range (~10 meters), restricting remote operation.
- Dependence on Connectivity:
The system relies on a stable Bluetooth connection; interruptions can affect performance.
- Limited Processing Capability:
Arduino has limited memory and processing power, restricting advanced functionalities.

- **Voice Recognition Accuracy:**
Accuracy depends on the mobile application and may be affected by noise or unclear speech.
- **Power Constraints:**
Battery life may limit continuous operation of the robot.
- **Lack of Internet Connectivity:**
The current model does not support remote access via the internet.

8.3 Future Scope

The project can be further enhanced with advanced features and technologies:

- **Integration with Wi-Fi/IoT Platforms:**
Replace Bluetooth with Wi-Fi modules (e.g., ESP8266) for remote control over the internet.
- **Cloud Integration:**
Use cloud databases for data storage, monitoring, and control.
- **Camera Integration:**
Add a camera for live video streaming and surveillance applications.
- **AI and Machine Learning:**
Implement intelligent decision-making and advanced voice recognition.
- **Mobile App Enhancement:**
Develop a customized application with improved UI/UX and additional features.
- **Autonomous Navigation:**
Upgrade the system to operate without human intervention using advanced sensors.
- **Increased Range and Efficiency:**
Use better communication modules and power systems for extended operation.

CHAPTER 9

REFERENCES

➤ REFERENCES

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